

1.3 Infinite Series

1.3.1 Sums of Infinite Series

Definition 1.3.1

An infinite series is an expression that can be written in the form

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \dots + u_k + \dots$$

The numbers $u_1, u_2, u_3, \dots$ are called the **terms** of the series.

Since it is impossible to add infinitely many numbers together directly, sums of infinite series are defined and computed by an indirect limiting process. To motivate the basic idea, consider the decimal

$$0.3333\dots \quad (1.8)$$

This can be viewed as the infinite series

$$0.3 + 0.03 + 0.003 + 0.0003 + \dots$$

or, equivalently,

$$\frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \dots \quad (1.9)$$

Since (1.8) is the decimal expansion of $\frac{1}{3}$, any reasonable definition for the sum of an infinite series should yield $\frac{1}{3}$ for the sum of ((1.9)). To obtain such a definition, consider the following sequence of (finite) sums:

$$\begin{aligned} s_1 &= \frac{3}{10} = 0.3 \\ s_2 &= \frac{3}{10} + \frac{3}{10^2} = 0.33 \\ s_3 &= \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} = 0.333 \\ s_4 &= \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} = 0.3333 \\ &\vdots \end{aligned}$$

The sequence of numbers $s_1, s_2, s_3, s_4, \dots$ can be viewed as a succession of approximations to the “sum” of the infinite series, which we want to be $\frac{1}{3}$. As we progress through the sequence, more and more terms of the infinite series are used, and the approximations get better and better, suggesting that the desired sum of $\frac{1}{3}$ might be the limit of this sequence of approximations. To see that this is so, we must calculate the limit

of the general term in the sequence of approximations, namely,

$$s_n = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \dots + \frac{3}{10^n} \quad (1.10)$$

The problem of calculating

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \left(\frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \dots + \frac{3}{10^n} \right)$$

is complicated by the fact that both the last term and the number of terms in the sum change with n . It is best to rewrite such limits in a closed form in which the number of terms does not vary, if possible. To do this, we multiply both sides of (1.10) by $\frac{1}{10}$ to obtain

$$\frac{1}{10} s_n = \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \dots + \frac{3}{10^n} + \frac{3}{10^{n+1}} \quad (1.11)$$

and then subtract (1.11) from (1.10) to obtain

$$s_n - \frac{1}{10} s_n = \frac{3}{10} - \frac{3}{10^{n+1}}$$

So $s_n = \frac{1}{3} \left(1 - \frac{1}{10^n} \right)$. It follows that

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \frac{1}{3} \left(1 - \frac{1}{10^n} \right) = \frac{1}{3}.$$

which we denote by writing

$$\frac{1}{3} = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \dots + \frac{3}{10^n} + \dots$$

Motivated by the preceding example, we are now ready to define the general concept of the “sum” of an infinite series

$$u_1 + u_2 + u_3 + \dots + u_k + \dots$$

We begin with some terminology: Let s_n denote the sum of the initial terms of the series, up to and including the term with index n . Thus,

$$s_1 = u_1$$

$$s_2 = u_1 + u_2$$

$$s_3 = u_1 + u_2 + u_3$$

$$\vdots$$

$$s_n = u_1 + u_2 + u_3 + \dots + u_n = \sum_{k=1}^n u_k$$

The number s_n is called the **n th partial sum** (ผลบวกย่อยอันดับที่ n หรือผลบวก n พจน์แรก) of the series and the sequence $\{s_n\}_{n=1}^{+\infty}$ is called the **sequence of partial sums**.

As n increases, the partial sum $s_n = u_1 + u_2 + \dots + u_n$ includes more and more terms of the series. Thus, if s_n tends toward a limit as $n \rightarrow +\infty$, it is reasonable to view this limit as the sum of *all* the terms in the series. This suggests the following definition.

Definition 1.3.2

Let $\{s_n\}$ be the sequence of partial sums of the series

$$u_1 + u_2 + u_3 + \dots + u_k + \dots$$

If the sequence $\{s_n\}$ converges to a limit S , then the series is said to **converge** (ลู่เข้า) to S , and S is called the **sum** (ผลบวก) of the series. We denote this by writing

$$S = \sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \dots + u_k + \dots$$

If the sequence of partial sums diverges, then the series is said to **diverge** (ลู่ออก). A divergent series has no sum.

Example 1.3.3

Determine whether the series

$$1 - 1 + 1 - 1 + 1 - 1 + \dots$$

converges or diverges. If it converges, find the sum.

1.3.2 Geometric Series

If the initial term of the series is a and each term is obtained by multiplying the preceding term by r , then the series has the form

$$\sum_{k=1}^{\infty} ar^{k-1} = a + ar + ar^2 + ar^3 + \dots + ar^{k-1} + \dots \quad (a \neq 0) \quad (1.12)$$

Such series are called **geometric series** (อนุกรมเรขาคณิต), and the number r is called the **common ratio** (อัตราส่วนร่วม) for the series.

The following theorem is the fundamental result on convergence of geometric series.

Theorem 1.3.4

A geometric series

$$\sum_{k=1}^{\infty} ar^{k-1} = a + ar + ar^2 + ar^3 + \dots + ar^{k-1} + \dots \quad (a \neq 0)$$

converges if $|r| < 1$ and diverges if $|r| \geq 1$. If the series converges, then the sum is

$$\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}$$

Proof.

- If $r = 1$, then the series is $a + a + a + \dots$ so the n th partial sum is $s_n = (n+1)a$ and

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (n+1)a = \pm\infty$$

(the sign depending on whether a is positive or negative). This proves divergence.

- If $r = -1$, then the series is $a - a + a - a + \dots$ so the sequence of partial sums is

$$a, 0, a, 0, a, 0, \dots$$

which diverges.

- Now let us consider the case where $|r| \neq 1$. The n th partial sum of the series is

$$s_n = a + ar + ar^2 + ar^3 + \dots + ar^n \quad (1.13)$$

Multiplying both sides of (1.13) by r yields

$$rs_n = ar + ar^2 + ar^3 + ar^4 + \dots + ar^n + ar^{n+1} \quad (1.14)$$

and subtracting (1.14) from (1.13) gives

$$s_n - rs_n = a - ar^{n+1}$$

Since $r \neq 1$ in the case we are considering, this can be rewritten as

$$s_n = \frac{a - ar^{n+1}}{1 - r} = \frac{a}{1 - r} (1 - r^{n+1}) \quad (1.15)$$

If $|r| < 1$, then $r^{n+1} \rightarrow 0$ as $n \rightarrow +\infty$, so $\{s_n\}$ converges. From (1.15)

$$\lim_{n \rightarrow +\infty} s_n = \frac{a}{1 - r}$$

If $|r| > 1$, then $r > 1$ or $r < -1$. In the case $r > 1$, r^{n+1} increases without bound as $n \rightarrow +\infty$, and in the case $r < -1$, r^{n+1} oscillates between positive and negative values that grow in magnitude, so $\{s_n\}$ diverges in both cases.

□

Example 1.3.5

In each part, determine whether the series converges, and if so find its sum.

(a) $\sum_{k=1}^{\infty} \frac{5}{4^k}$

(b) $\sum_{k=1}^{\infty} 3^{2k} 5^{1-k}$

Example 1.3.6

In each part, find all values of x for which the series converges, and find the sum of the series for those values of x .

(a) $\sum_{k=0}^{\infty} x^k$

(b) $3 - \frac{3x}{2} + \frac{3x^2}{4} - \frac{3x^3}{8} + \dots + \frac{3(-1)^k}{2^k} x^k + \dots$

1.3.3 Telescoping Sums

The sum is called a **telescoping sum** if, in simplifying the sum, one term in each parenthetical expression cancels one term in the next parenthetical expression, until the entire sum collapses (like a folding telescope) into just two terms.

Example 1.3.7

Determine whether the series

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots$$

converges or diverges. If it converges, find the sum.

1.3.4 Harmonic Series

One of the most important of all diverging series is the **harmonic series**,

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

which arises in connection with the overtones produced by a vibrating musical string. It is not immediately evident that this series diverges. However, the divergence will become apparent when we examine the partial sums in detail. Because the terms in the series are all positive, the partial sums

$$s_1, \quad s_2 = 1 + \frac{1}{2}, \quad s_3 = 1 + \frac{1}{2} + \frac{1}{3}, \dots$$

form a strictly increasing sequence

$$s_1 < s_2 < s_3 < \dots < s_n < \dots$$

Consider

$$\begin{aligned} s_2 &= 1 + \frac{1}{2} > \frac{1}{2} + \frac{1}{2} = \frac{2}{2} \\ s_4 &= s_2 + \frac{1}{3} + \frac{1}{4} > s_2 + \left(\frac{1}{4} + \frac{1}{4}\right) = s_2 + \frac{1}{2} > \frac{3}{2} \\ s_8 &= s_4 + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > s_4 + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) = s_4 + \frac{1}{2} > \frac{4}{2} \\ &\vdots \\ s_{2^n} &> \frac{n+1}{2} \end{aligned}$$

If M is any constant, we can find a positive integer n such that $\frac{n+1}{2} > M$. But for this n

$$s_{2^n} > \frac{n+1}{2} > M$$

so that no constant M is greater than or equal to *every* partial sum of the harmonic series. This proves divergence.

This divergence proof, which predates the discovery of calculus, is due to a French bishop and teacher, Nicole Oresme (1323–1382). This series eventually attracted the interest of Johann and Jakob Bernoulli and led them to begin thinking about the general concept of convergence, which was a new idea at that time.

1.3.5 Exercises

1. Determine whether the series converges, and if so find its sum.

1.1. $1 + \frac{3}{2} + \frac{9}{4} + \frac{27}{8} + \dots$

1.2. $3 + \frac{5}{2} + \frac{7}{3} + \frac{9}{4} + \dots$

1.3. $3 + \sqrt{3} + \sqrt[3]{3} + \sqrt[4]{3} + \dots$

1.4. $\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \frac{1}{4 \cdot 6} + \dots$

1.5. $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots$

1.6. $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots$

1.7. $\sum_{n=1}^{\infty} (\ln 3)^n$

1.8. $\sum_{n=1}^{\infty} \frac{2^n + 3}{5^n}$

1.9. $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2}$

1.10. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + \sqrt{n-1}}$

1.11. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + \sqrt{n^2-1}}$

1.12. $\sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right)$

1.13. $\sum_{n=1}^{\infty} \left(\frac{4}{n^2 + 4n + 3} - 9^{2-n} 4^{n+1} \right)$

2. Find all values of x for which the series converges.

2.1. $\sum_{n=1}^{\infty} (-1)^n x^{2n}$

2.2. $\sum_{n=1}^{\infty} (\ln x)^n$

3. Show that for all real values of x

$$\sin x - \frac{1}{2} \sin^2 x + \frac{1}{4} \sin^3 x - \frac{1}{8} \sin^4 x + \dots = \frac{2 \sin x}{2 + \sin x}.$$

4. Find the solution set of the equation $1 + e^x + e^{2x} + e^{3x} + \dots = 9$.

5. Let x be a real number such that $|x| < 1$. Find all values of x for which the series

$$1 + (1+x)\frac{1}{2} + (1+x+x^2)\left(\frac{1}{2}\right)^2 + (1+x+x^2+x^3)\left(\frac{1}{2}\right)^3 + \dots$$

converges to $\frac{16}{7}$.

6. Determine whether the series

$$\log_2 x - (\log_2 x)^2 + (\log_2 x)^3 - (\log_2 x)^4 + \dots$$

converges or diverges if the series $1 + \frac{2^x}{1+2^x} + \frac{2^{2x}}{(1+2^x)^2} + \frac{2^{3x}}{(1+2^x)^3} + \dots$ converges to 9.

7. Let θ be a real number. Find the sum of the series

$$\sin \theta + \frac{\sin 2\theta}{\pi} + \frac{\sin 3\theta}{\pi^2} + \dots + \frac{\sin n\theta}{\pi^{n-1}} + \dots$$

8. Determine whether the series converges, and if so find its sum.

8.1. $1 + \frac{3}{4} + \frac{7}{16} + \frac{15}{64} + \dots$

8.2. $1 + \frac{5}{3} + \frac{12}{3^2} + \frac{22}{3^3} + \frac{35}{3^4} + \dots$

8.3. $\frac{1}{4} \log 2 + \frac{1}{8} \log 4 + \frac{1}{16} \log 8 + \frac{1}{32} \log 16 + \dots$

9. The great Swiss mathematician Leonhard Euler sometimes reached incorrect conclusions in his pioneering work on infinite series. For example, Euler deduced that

$$\frac{1}{2} = 1 - 1 + 1 - 1 + \dots$$

and

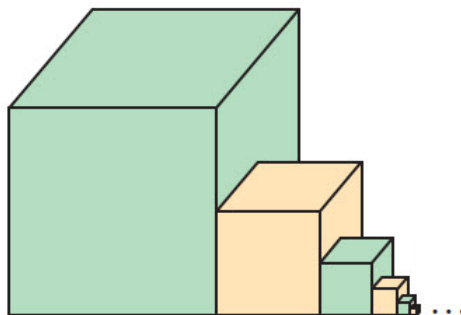
$$-1 = 1 + 2 + 4 + 8 + \dots$$

by substituting $x = -1$ and $x = 2$ in the formula

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

What was the problem with his reasoning?

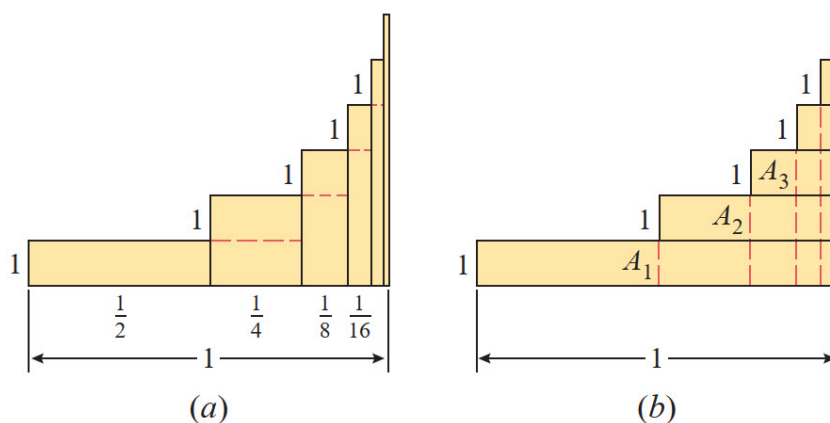
10. A ball is dropped from a height of 10 m. Each time it strikes the ground it bounces vertically to a height that is $3/4$ of the preceding height. Find the total distance the ball will travel if it is assumed to bounce infinitely often.
11. The accompanying figure shows an “infinite staircase” constructed from cubes. Find the total volume of the staircase, given that the largest cube has a side of length 1 and each successive cube has a side whose length is half that of the preceding cube.



12. In his *Treatise on the Configurations of Qualities and Motions* (written in the 1350s), the French Bishop of Lisieux, Nicole Oresme, used a geometric method to find the sum of the series

$$\sum_{k=1}^{\infty} \frac{k}{2^k} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$$

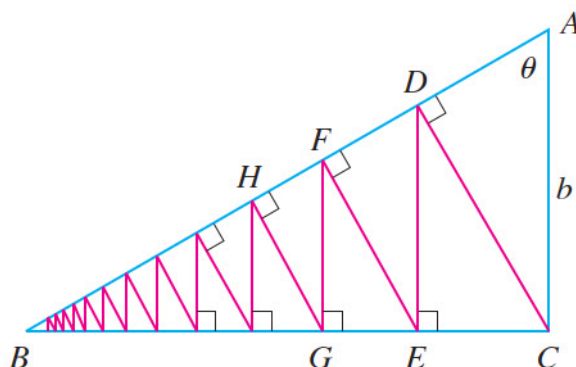
In part (a) of the accompanying figure, each term in the series is represented by the area of a rectangle, and in part (b) the configuration in part (a) has been divided into rectangles with areas $A_1, A_2, A_3, \dots$. Find the sum $A_1 + A_2 + A_3 + \dots$.



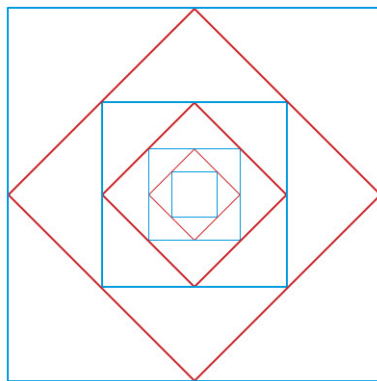
13. A right triangle ABC is given with $\angle A = \theta$ and $|AC| = b$. CD is drawn perpendicular to AB , DE is drawn perpendicular to BC , $EF \perp AB$, and this process is continued indefinitely, as shown in the figure. Find the total length of all the perpendiculars

$$|CD| + |DE| + |EF| + |FG| + \dots$$

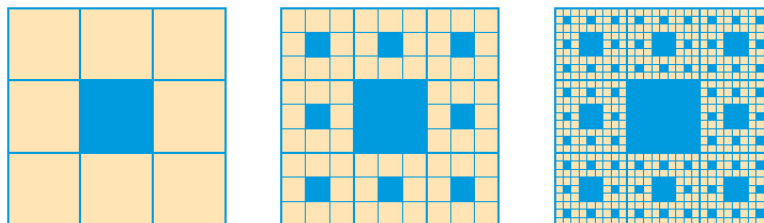
in terms of b and θ .



14. The midpoints of the sides of a square of side 1 are joined to form a new square. This procedure is repeated for each new square. (See the figure.)



- 14.1. Find the sum of the areas of all the squares.
- 14.2. Find the sum of the perimeters of all the squares.
15. A yellow square of side 1 is divided into nine smaller squares, and the middle square is colored blue as shown in the figure. Each of the smaller yellow squares is in turn divided into nine squares, and each middle square is colored blue. If this process is continued indefinitely, what is the total area that is colored blue?



16. (a) A sequence $\{a_n\}$ is defined recursively by the equation

$$a_n = \frac{1}{2}(a_{n-1} + a_{n-2}) \quad \text{for } n \geq 3$$

where a_1 and a_2 can be any real numbers. Experiment with various values of a_1 and a_2 and use your calculator to guess the limit of the sequence.

- (b) Find $\lim_{n \rightarrow +\infty} a_n$ in terms of a_1 and a_2 .

Hint: Express $a_{n+1} - a_n$ in terms of $a_2 - a_1$ and sum a series.